

Hotel WhatsApp Bot — Technical Requirements

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2026-05-11

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Python + VPS + Supabase Stack

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Python + VPS + Supabase Stack

Version: 1.0

Date: 2026-05-11

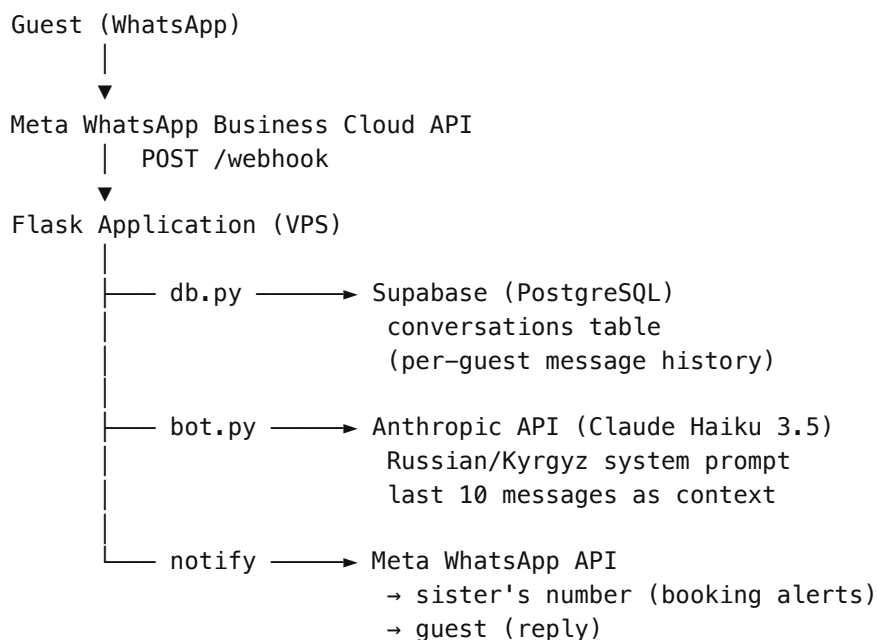
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Status: Draft — pending review

1. Overview

A Python-based WhatsApp AI assistant for a mini-hotel at Lake Issyk-Kul, Kyrgyzstan. Guests send WhatsApp messages in Russian or Kyrgyz; the bot answers FAQs automatically using Claude Haiku and notifies the owner's sister when a booking request is detected. All conversation history is stored in Supabase so the bot remembers context across messages and the owner can review all chats.

2. System Architecture



3. Infrastructure

3.1 VPS

Parameter	Specification
Provider	Hetzner Cloud (recommended) or any VPS
Plan	CX22 — 2 vCPU, 4GB RAM, 40GB SSD
Cost	€4–6/month
OS	Ubuntu 22.04 LTS
Region	EU (Falkenstein or Helsinki — closest to Kyrgyzstan)
Public IP	Required (static, for webhook)

Required services on VPS:

- Python 3.11+
- gunicorn (WSGI server)
- nginx (reverse proxy, SSL termination)
- systemd service (auto-restart on crash/reboot)
- Certbot (free Let's Encrypt SSL — required by Meta for webhooks)

3.2 DOMAIN

A domain name is required for HTTPS (Meta rejects non-HTTPS webhook URLs).

Options:

- Buy a cheap domain (~\$10/year) from Namecheap or Cloudflare
- Point an A record to the VPS IP
- Use Certbot to get a free SSL certificate

Alternative (no domain): use Cloudflare Tunnel (free) to expose the Flask app over HTTPS without a domain.

3.3 SUPABASE

Parameter	Specification
Plan	Free tier (500MB DB, 2GB bandwidth)
Region	Europe West (closest to Kyrgyzstan)
Auth	Service Role key (server-side only, never exposed)
Interface	Supabase Dashboard (owner can view all chats)

4. File Structure

```

hotel-bot/
├── app.py           ← Flask app: webhook handler, routing
├── bot.py           ← Claude Haiku call, system prompt, reply logic
├── db.py            ← Supabase: read/write conversation history
├── notify.py        ← Send WhatsApp notification to sister
├── system-prompt.txt ← Russian/Kyrgyz hotel prompt (owner edits this)
├── requirements.txt ← Python dependencies
├── .env.example     ← Environment variable template (no secrets)
├── .env             ← Actual secrets (never committed to git)
├── .gitignore       ← Excludes .env, __pycache__, etc.
├── deploy/
│   ├── hotel-bot.service ← systemd unit file
│   ├── nginx.conf        ← nginx reverse proxy config
│   └── deploy.md         ← Step-by-step VPS setup guide
├── sql/
│   └── schema.sql        ← Supabase table creation SQL
├── test-messages.txt    ← Manual test script
└── docs/
    ├── meta-setup.md    ← Meta WhatsApp Business API setup guide
    └── tech-requirements-python.md ← This document

```

5. Database Schema (Supabase)

TABLE: `CONVERSATIONS`

```
CREATE TABLE conversations (
  id          BIGSERIAL PRIMARY KEY,
  phone_number TEXT NOT NULL,
  messages    JSONB NOT NULL DEFAULT '[]',
  updated_at  TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE UNIQUE INDEX conversations_phone_number_idx
ON conversations (phone_number);
```

`messages` **JSONB structure** (array of message objects):

```
[
  { "role": "user",      "content": "Здравствуйте, сколько стоит номер?" },
  { "role": "assistant", "content": "Добрый день! Стандартный номер – 2500 сом/ночь." },
  { "role": "user",      "content": "Хочу забронировать" }
]
```

Retention policy: Keep last 20 messages per guest. Older messages are trimmed on each write to keep the array bounded.

6. API Integrations

6.1 META WHATSAPP BUSINESS CLOUD API

Parameter	Detail
Endpoint	<code>https://graph.facebook.com/v19.0/{PHONE_NUMBER_ID}/messages</code>
Auth	Bearer token (permanent system user token)
Webhook path	<code>POST /webhook</code>
Webhook verification	<code>GET /webhook</code> — verify token handshake
Inbound event	<code>messages</code> <code>subscription</code>
Message type handled	<code>text</code> only (v1 — no media)
Rate limit	1,000 messages/second (not a concern at hotel scale)

Webhook payload (inbound message):

```
{
  "entry": [{
    "changes": [{
      "value": {
        "messages": [{
          "from": "996XXXXXXXXX",
          "text": { "body": "Здравствуйте" },
          "type": "text"
        }]
      }
    }]
  }]
}
```

Outbound send (reply to guest):


```
{
  "messaging_product": "whatsapp",
  "to": "996XXXXXXXXX",
  "type": "text",
  "text": { "body": "Добрый день! Чем могу помочь?" }
}
```

6.2 ANTHROPIC CLAUDE HAIKU 3.5

Parameter	Detail
Model	claude-haiku-4-5-20251001
Max tokens	400 (sufficient for hotel FAQ replies)
Context window	Last 10 messages from Supabase + system prompt
System prompt	Contents of system-prompt.txt (loaded at startup)
Languages	Russian (primary), Kyrgyz, English
Cost	~\$0.80/1M input tokens, ~\$4.00/1M output tokens
Est. monthly cost	~\$1–2/month at 1,000 messages

6.3 SUPABASE PYTHON CLIENT

Parameter	Detail
Library	supabase (official Python client)
Auth	SUPABASE_URL + SUPABASE_SERVICE_KEY
Operations	upsert (write), select (read) on conversations table
Connection	HTTP (no persistent connection needed at this scale)

7. Environment Variables

All secrets stored in .env on the VPS. Never committed to git.

```
# Meta WhatsApp
WHATSAPP_ACCESS_TOKEN=      # Permanent system user token
WHATSAPP_PHONE_NUMBER_ID=   # Phone Number ID from Meta Developer Console
WHATSAPP_VERIFY_TOKEN=hotel-bot-verify-2026

# Sister notification
SISTER_PHONE_NUMBER=996XXXXXXX # No + sign, no spaces

# Anthropic
ANTHROPIC_API_KEY=          # From console.anthropic.com

# Supabase
SUPABASE_URL=               # https://xxxx.supabase.co
SUPABASE_SERVICE_KEY=       # Service role key (not anon key)

# App
FLASK_ENV=production
PORT=8000
```

8. Application Logic

8.1 WEBHOOK HANDLER (`APP.PY`)

```
POST /webhook
1. Verify request is from Meta (check X-Hub-Signature-256 header)
2. Extract: sender phone number, message text, message type
3. Ignore non-text messages (images, voice notes, etc.) – return 200 OK
4. Ignore messages from the bot's own number (echo prevention)
5. Pass to bot.py → get reply text
6. Send reply to guest via Meta API
7. Return 200 OK to Meta (within 5 seconds – Meta retries if no 200)

GET /webhook
1. Verify hub.verify_token matches WHATSAPP_VERIFY_TOKEN
2. Return hub.challenge if valid
```

8.2 BOT LOGIC (`BOT.PY`)

```

handle_message(phone_number, message_text):
    1. Load conversation history from Supabase (last 10 messages)
    2. Append new user message to history
    3. Call Claude Haiku with system_prompt + history
    4. Append assistant reply to history
    5. Trim history to last 20 messages
    6. Save updated history to Supabase (upsert)
    7. Check for booking intent keywords in message_text:
        ["забронировать", "бронь", "свободен", "book", "reserve", "хочу номер"]
    8. If booking intent → call notify.py with phone_number + message_text + reply
    9. Return reply text

```

8.3 BOOKING INTENT DETECTION

Simple keyword match on the incoming message (case-insensitive). When triggered:

- Send a WhatsApp message to the sister's number with:
 - Guest phone number
 - Guest's message
 - Bot's reply

This is a v1 implementation. A future version can use Claude to return structured JSON with `intent: booking` for more accurate detection.

8.4 SESSION MEMORY (`DB.PY`)

```

get_history(phone_number) → list[dict]
    SELECT messages FROM conversations WHERE phone_number = $1
    Returns [] if no row exists

save_history(phone_number, messages: list[dict])
    UPSERT into conversations (phone_number, messages, updated_at)
    Trim messages to last 20 before saving

```

9. Security Requirements

Requirement	Implementation
Webhook authenticity	Verify <code>X-Hub-Signature-256</code> HMAC on every POST
Secrets management	<code>.env</code> file on VPS, never in git
HTTPS only	nginx with Let's Encrypt certificate
Supabase key	Service role key used server-side only
No payment data	System prompt explicitly prohibits bot from sharing payment details
Input sanitization	Strip and truncate incoming message text (max 4,000 chars)

10. Deployment

10.1 VPS SETUP (ONE-TIME)

```
# 1. Update system
apt update && apt upgrade -y

# 2. Install Python and tools
apt install python3.11 python3.11-venv python3-pip nginx certbot python3-certbot-nginx -y

# 3. Create app user
useradd -m -s /bin/bash hotelbot

# 4. Clone repo and install deps
su - hotelbot
git clone https://github.com/ZeekrBaha/issykkul-hotel-whatsapp-bot.git
cd issykkul-hotel-whatsapp-bot
python3.11 -m venv venv
source venv/bin/activate
pip install -r requirements.txt

# 5. Create .env with real values
cp .env.example .env
nano .env

# 6. Install systemd service
cp deploy/hotel-bot.service /etc/systemd/system/
systemctl enable hotel-bot
systemctl start hotel-bot

# 7. Configure nginx
cp deploy/nginx.conf /etc/nginx/sites-available/hotel-bot
ln -s /etc/nginx/sites-available/hotel-bot /etc/nginx/sites-enabled/
certbot --nginx -d yourdomain.com
systemctl reload nginx
```

10.2 UPDATING THE BOT

```
# On VPS
su - hotelbot
cd issykkul-hotel-whatsapp-bot
git pull
source venv/bin/activate
pip install -r requirements.txt # if deps changed
systemctl restart hotel-bot
```

10.3 UPDATING THE SYSTEM PROMPT (NO REDEPLOY NEEDED)

Edit `system-prompt.txt` on the VPS and restart the service:

```
nano system-prompt.txt
systemctl restart hotel-bot
```

11. Python Dependencies (`requirements.txt`)

```
flask==3.0.3
gunicorn==21.2.0
anthropic==0.25.0
supabase==2.4.2
python-dotenv==1.0.1
requests==2.31.0
```

12. Cost Summary

Item	Monthly Cost
Hetzner VPS CX22	€4 (\$4.50)
Domain (optional)	~\$1 (amortized)
Supabase	\$0 (free tier)
Meta WhatsApp service conversations	\$0 (free for inbound)
Anthropic Claude Haiku (1,000 msg)	~\$2
Total	~\$7–8/month

At 5,000 messages/month (busier season): ~\$15–20/month.

13. Functional Requirements

ID	Requirement
FR-1	Bot replies to WhatsApp messages in Russian within 5 seconds
FR-2	Bot answers FAQs: prices, room types, check-in/out, amenities, directions
FR-3	Bot detects booking intent via keyword matching
FR-4	Bot collects: guest name, check-in date, check-out date, number of guests
FR-5	Bot notifies sister via WhatsApp when booking intent is detected
FR-6	Bot responds in Kyrgyz if guest writes in Kyrgyz
FR-7	Bot responds in English if guest writes in English
FR-8	Bot never shares payment details
FR-9	Conversation history persists across messages (Supabase)
FR-10	System prompt is editable without redeploying code

14. Non-Functional Requirements

ID	Requirement
NFR-1	Uptime: 99%+ (systemd auto-restart)
NFR-2	Response time: under 5 seconds end-to-end
NFR-3	Cost: under \$20/month at peak season volume
NFR-4	All secrets in environment variables, never in code
NFR-5	HTTPS required (Meta webhook requirement)
NFR-6	Webhook signature verification on every request

15. Out of Scope (v1)

- Voice/phone call handling (Phase 2 — VAPI + Zadarma)

- Media messages (images, voice notes, documents)
- Booking calendar / availability database
- Payment processing
- Admin dashboard beyond Supabase table viewer
- Multi-hotel support